AI Automation Opportunities for a Telecommunication NOC Engineer

1. Automated Site-Down Notification Escalation (Recommended)

Problem

As a Network Operations Center (NOC) Engineer, one of the most repetitive tasks is monitoring site-down alarms. When a telecom site goes down, the monitoring system sends an alarm, and the NOC engineer must manually:

- Check the alarm.
- Confirm the affected site.
- Identify the engineer responsible for that region.
- Send an email or WhatsApp message.
- Follow up until the issue is acknowledged.

During major outages involving many sites, this manual process becomes slow and increases the response time.

AI Automation Solution

An AI automation workflow can monitor a dedicated email folder where site-down alarms are received.

Trigger: A new site-down notification arrives in the dedicated email folder.

The automation will:

1. Read the alarm email.
2. Extract the site ID, site name, location, and time of the outage.
3. Use AI to summarize the incident and determine its priority.
4. Identify the responsible field engineer based on the site location.
5. Automatically send an escalation message via Microsoft Teams, Outlook, SMS, or WhatsApp.
6. Log the incident into a ticketing system such as ServiceNow or Jira.
7. Notify the NOC supervisor if the alarm is not acknowledged within a defined period.

Possible Tools

- Microsoft Power Automate
- OpenAI (ChatGPT API)
- Outlook
- Microsoft Teams
- ServiceNow or Jira
- Excel or SharePoint (Engineer assignment list)

Simple Workflow

Site goes down

Monitoring tool sends alarm email

Power Automate monitors dedicated email folder

AI extracts site details and priority

Find assigned field engineer

Automatically send escalation message

Create incident ticket

Monitor acknowledgement

Escalate to supervisor if no response

Business Impact

- Faster fault escalation.

- Reduced response time (MTTR).
- Improved SLA compliance.
- Eliminates manual notification.
- Ensures every site-down alarm is attended to promptly.
- Allows NOC engineers to focus on monitoring instead of repetitive communication.

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2. AI Validation of Minor Power Alarms (False Alarm Detection)

Problem

NOC engineers receive many minor alarms from telecom sites every day.

Before escalating these alarms, engineers usually need to verify whether the alarm is genuine or a false alarm by checking live site parameters on the monitoring system.

For example, when a power-related alarm is received, the engineer manually checks:

- Site voltage.
- Battery level.
- Rectifier status.
- Generator status.
- Whether the site voltage is below **48V**.
- Other site information displayed on the monitoring platform.

This verification process is repeated hundreds of times every day.

AI Automation Solution

An AI assistant can automatically validate incoming minor alarms using the live data from the monitoring platform.

Trigger: A new minor alarm appears on the monitoring tool.

The automation will:

1. Read the alarm details.
2. Retrieve the site's live operating parameters.
3. Check whether the site voltage is below **48V**.

4. Analyze other parameters such as battery level and power source.
5. Determine whether the alarm is likely genuine or a false alarm.
6. Recommend the next action:
    - Ignore false alarm.
    - Continue monitoring.
    - Escalate immediately.

If escalation is required, the automation can automatically create a ticket and notify the responsible engineer.

Possible Tools

- Microsoft Power Automate
- Python
- OpenAI API
- Network Monitoring System (OWS, Galooli RMS, Huawei U2000, etc.)
- ServiceNow
- Microsoft Teams

Simple Workflow

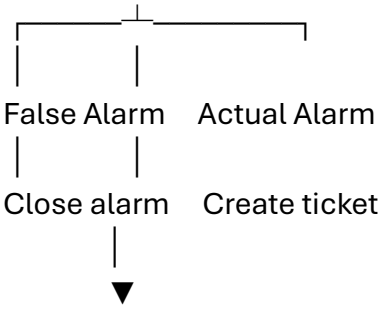
Minor alarm generated

Automation detects alarm

Retrieve live site information

Check voltage
(Below 48V?)

AI analyzes site condition



Business Impact

- Reduces unnecessary escalations.
- Eliminates time spent manually checking site parameters.
- Improves alarm accuracy.
- Reduces false dispatches of field engineers.
- Speeds up decision-making.
- Allows engineers to focus on genuine network faults.